

OFFICE AUTOMATION TOOLS

Unit-I: DOS Commands and Windows 2007

Comprehensive Study Notes

1. Introduction to DOS

1.1 What is DOS?

DOS (Disk Operating System) is a **command-line operating system** that was widely used in early personal computers. It provides a **text-based interface** where users type commands to perform operations such as managing files, running programs, and organizing data.

1.2 Types of DOS Commands

DOS commands are classified into two main categories:

Internal Commands

Internal commands are **built into** the command interpreter (COMMAND.COM). They are loaded into memory when the system starts and are always available.

Examples: DIR, DATE, TIME, CLS, CD, RD, MD, COPY, DEL, TYPE, REN, VER, VOL, PROMPT, PATH, ECHO

External Commands

External commands are **separate program files** stored on the disk. They need to be present in the current directory or in the path to be executed.

Examples: ATTRIB, CHKDSK, DISKCOPY, DISKCOMP, XCOPY, TREE, DELTREE, DOSKEY, FORMAT, FIND, SORT, FDISK, MORE, SYS

Feature	Internal Commands	External Commands
Location	Part of COMMAND.COM	Separate .COM or .EXE files
Availability	Loaded into RAM when system boots	Need to be present on disk
Examples	DIR, COPY, DEL	FORMAT, CHKDSK, XCOPY

2. Internal DOS Commands

2.1 DIR - Directory Listing

Purpose: Displays a list of files and subdirectories in a directory.

Syntax: DIR [drive:][path][filename] [/P]/[W]/[OD]/[S]

Option	Description
/P	Pause after each screenful
/W	Wide listing format (across the screen)
/OD	List by date order
/S	List files in all subdirectories
/OGNE	List by directory, name, and extension

```
DIR
DIR C:\DOS
DIR /P
DIR /W
```

2.2 DATE - Display/Set Date

Purpose: Displays or changes the system date.

Syntax: DATE [mm-dd-yy]

```
DATE 12-25-2025
```

2.3 TIME - Display/Set Time

Purpose: Displays or sets the system time.

Syntax: TIME [hh:mm:ss]

```
TIME 14:30:00
```

2.4 CLS - Clear Screen

Purpose: Clears the screen and places the cursor at the top-left corner.

Syntax: CLS

```
CLS
```

2.5 CD / CHDIR - Change Directory

Purpose: Changes the current working directory.

Syntax: CD [drive:][path]

Command	Action
CD \	Move to root directory
CD ..	Move to parent directory
CD \BAKER\MYFILES	Move to specific directory

```
CD \  
CD ..  
CD MYFOLDER
```

2.6 RD / RMDIR - Remove Directory

Purpose: Removes (deletes) an **empty** directory.

Syntax: RD [drive:]path

```
RD TEST
```

Note: Directory must be empty before removal.

2.7 MD / MKDIR - Make Directory

Purpose: Creates a new directory.

Syntax: MD [drive:]path

```
MD \ABLE  
MD MYFOLDER
```

2.8 PATH - Set Search Path

Purpose: Displays or sets the search path for executable files.

Syntax: PATH [drive:]path

```
PATH C:\DOS;C:\WINDOWS
```

2.9 TYPE - Display File Contents

Purpose: Displays the contents of a text file on the screen.

Syntax: TYPE [drive:][path]filename

```
TYPE MYLETTER.TXT
```

2.10 DEL / ERASE - Delete Files

Purpose: Deletes one or more files.

Syntax: DEL [drive:][path]filename

```
DEL MYFILE.LTR  
DEL *.TMP
```

2.11 COPY - Copy Files

Purpose: Copies one or more files from one location to another.

Syntax: COPY [source] [destination]

```
COPY MYFILE.TXT YOURFILE.TXT  
COPY C:\DOCS\*.TXT A:
```

2.12 REN / RENAME - Rename Files

Purpose: Changes the name of a file or directory.

Syntax: REN [old_name] [new_name]

```
RENAME MYTEST.TST MYOLD.TST
```

2.13 PROMPT - Change Command Prompt

Purpose: Changes the appearance of the DOS command prompt.

Syntax: PROMPT [text]

Code	Meaning
\$P	Current drive and path
\$G	Greater than sign (>)
\$D	Current date
\$T	Current time

```
PROMPT $P$G  
PROMPT Hello$G
```

2.14 VOL - Display Volume Label

Purpose: Displays the disk volume label and serial number.

Syntax: VOL [drive:]

```
VOL C:
```

2.15 VER - Display DOS Version

Purpose: Displays the version of DOS being used.

Syntax: VER

```
VER
```

2.16 ECHO - Display Messages

Purpose: Displays messages or turns on/off the display of commands in batch files.

Syntax: ECHO [on|off|message]

```
ECHO Hello World  
ECHO OFF
```

3. External DOS Commands

3.1 ATTRIB - Change File Attributes

Purpose: Displays or changes file attributes (Read-Only, Archive, System, Hidden).

Syntax: ATTRIB [+R|-R] [+A|-A] [+S|-S] [+H|-H] [filename]

Attribute	Description
+R / -R	Set/Remove Read-Only
+A / -A	Set/Remove Archive
+S / -S	Set/Remove System
+H / -H	Set/Remove Hidden

```
ATTRIB +R MYFILE
ATTRIB -R MYFILE
```

3.2 CHKDSK - Check Disk

Purpose: Checks a disk and provides a file and memory status report.

Syntax: CHKDSK [drive:] [/F] [/V]

Option	Description
/F	Fixes errors on the disk
/V	Displays the name of each file as it is checked

```
CHKDSK C:
```


3.3 DISKCOPY - Copy Floppy Disk

Purpose: Duplicates a floppy disk to another disk of the same type and size.

Syntax: DISKCOPY [source_drive:] [destination_drive:]

```
DISKCOPY A: A:
```

3.4 DISKCOMP - Compare Floppy Disks

Purpose: Compares the contents of two floppy disks.

Syntax: DISKCOMP [drive1:] [drive2:]

```
DISKCOMP A: B:
```

3.5 XCOPY - Extended Copy

Purpose: Copies files and directories, including subdirectories.

Syntax: XCOPY [source] [destination] [/S] [/E]

Option	Description
/S	Copies directories and subdirectories
/E	Copies subdirectories even if empty

```
XCOPY C:\DOCS A: /S
```

3.6 TREE - Display Directory Structure

Purpose: Displays the directory structure of a disk or path.

Syntax: TREE [drive:][path] [/F]

Option	Description
/F	Displays files along with directories

```
TREE C:\
TREE /F
```

3.7 DELTREE - Delete Directory Tree

Purpose: Deletes a directory and all its subdirectories and files.

Warning: This is a **dangerous command** as it permanently deletes everything.

Syntax: DELTREE [drive:]path

```
DELTREE C:\OLD
```

3.8 DOSKEY - Command History

Purpose: Loads the DOSKey program which allows recalling and editing previous commands.

Syntax: DOSKEY [install]

```
DOSKEY
```

Key Functions:

- Up Arrow: Recall previous command
- Down Arrow: Recall next command
- F7: List command history

3.9 FORMAT - Format Disk

Purpose: Prepares a disk to accept data.

Syntax: FORMAT [drive:] [/S] [/Q]

Option	Description
/S	Creates a system/bootable disk
/Q	Quick format

```
FORMAT A:  
FORMAT A: /S
```

3.10 FIND - Find String

Purpose: Searches for a specific text string in one or more files.

Syntax: FIND ["string"] [files]

Option	Description
/V	Displays lines not containing the string
/C	Counts lines containing the string
/I	Case-insensitive search

```
FIND "error" LOG.TXT
```

3.11 SORT - Sort Data

Purpose: Sorts text data alphabetically or numerically.

Syntax: SORT [/R] [/N] [input_file] [output_file]

Option	Description
/R	Reverse sort order
/N	Numeric sort

```
SORT < DATA.TXT
```

3.12 FDISK - Fixed Disk Setup

Purpose: Configures hard disk partitions.

Syntax: FDISK

```
FDISK
```

Functions:

- Create partitions
- Delete partitions
- Set active partition

3.13 MORE - Page by Page Display

Purpose: Displays data one screen at a time.

Syntax: MORE < [file]

```
TYPE LARGE.TXT | MORE  
DIR | MORE
```

3.14 SYS - Transfer System Files

Purpose: Copies DOS system files and command interpreter to a disk.

Syntax: SYS [drive:]

```
SYS A:
```

4. Concept of Files and Directories

4.1 Files

A **file** is a collection of related data stored with a unique name.

File Naming Rules in DOS

Rule	Description
Filename	1 to 8 characters
Extension	0 to 3 characters (optional)
Format	filename.extension
Valid Characters	A-Z, 0-9, _ ^ \$ ~ ! # % & - { } () @
Invalid Characters	spaces, < > . , ; : " / \ ? *

Examples:

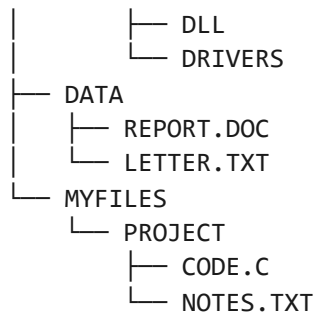
- REPORT.DOC
- LETTER.TXT
- MYFILE

4.2 Directories

A **directory** (or folder) is a container that holds files and other directories.

Directory Structure

```
ROOT DIRECTORY
├── DOS
│   ├── COMMAND.COM
│   └── FORMAT.COM
├── WINDOWS
│   ├── WIN.COM
│   └── SYSTEM
```



Path Names

Type	Description	Example
Absolute Path	Full path from root	C:\DOS\COMMAND.COM
Relative Path	Path from current directory	..\DATA\REPORT.DOC

5. Wildcard Characters

5.1 What are Wildcard Characters?

Wildcards are special characters that represent one or more characters in a filename. They are used to perform operations on multiple files at once.

5.2 Types of Wildcards

Wildcard	Description	Example
*	Represents any number of characters	*.TXT = all .TXT files
?	Represents a single character	FILE?.TXT = FILE1.TXT, FILE2.TXT

5.3 Wildcard Examples

Command	Files Matched
*.DOC	All .DOC files
REPORT.*	All files named REPORT
FILE?.TXT	FILE1.TXT, FILE2.TXT, etc.
.	All files
A*.EXE	Files starting with A ending .EXE
???.TXT	Files with 3-letter names ending .TXT

5.4 Commands Using Wildcards

```
COPY *.TXT A:  
DEL *.TMP  
DIR *.EXE  
DEL *.*
```


6. Redirection Operators

6.1 What are Redirection Operators?

Redirection operators control where input comes from and where output goes.

6.2 Redirection Operators Table

Operator	Description	Example
>	Redirects output to a file (overwrites)	DIR > LIST.TXT
>>	Redirects output to a file (appends)	DIR >> LIST.TXT
<	Takes input from a file	SORT < DATA.TXT
	Pipes output to another command	DIR MORE

6.3 Examples

Output Redirection (>)

```
DIR > FILELIST.TXT  
ECHO Hello > MESSAGE.TXT
```

Output Redirection (>>)

```
DIR >> FILELIST.TXT
```

Input Redirection (<)

```
SORT < UNSORTED.TXT
```

Piping (|)

DIR | MORE

TYPE FILE.TXT | FIND "error"

7. Windows 2007

7.1 What is Windows 2007?

Windows 2007 is a **graphical user interface (GUI) operating system** developed by Microsoft. It provides a user-friendly interface with icons, windows, and menus instead of command-line input.

7.2 Benefits of Windows 2007

Benefit	Description
User-Friendly	GUI makes it easy to use
Multi-Tasking	Run multiple applications simultaneously
Plug and Play	Automatic hardware detection
Visual Interface	Icons, windows, and menus
File Management	Easy file organization
Networking	Built-in network support
Security	User accounts and permissions
Multimedia	Built-in support for audio and video

7.3 Features of Windows 2007

Feature	Description
Desktop	Main screen with icons and taskbar
Start Menu	Access to programs and settings

Taskbar	Shows running applications
Windows Explorer	File management
Control Panel	System settings
Search	Find files and programs
Gadgets	Desktop widgets
Aero Interface	Visual effects (window transparency)

7.4 Uses of Windows 2007

Use	Description
Personal Computing	Daily tasks, internet, entertainment
Office Work	Document creation, spreadsheets
Education	Learning and research
Business	Professional applications
Communication	Email, messaging, video calls
Design	Graphics, multimedia

8. Control Panel

8.1 What is Control Panel?

Control Panel is a **system settings management tool** that allows users to view and change system settings.

8.2 Control Panel Categories

Category	Settings
Appearance and Personalization	Wallpaper, themes, fonts
Clock, Language, and Region	Date, time, language
Hardware and Sound	Printers, mouse, sound
Network and Internet	Network connections, internet settings
Programs	Add/Remove programs
System and Security	System information, firewall, updates
User Accounts	User creation, password
Ease of Access	Accessibility options

8.3 Common Control Panel Tasks

Task	Location
Change Wallpaper	Appearance and Personalization
Install/Uninstall Programs	Programs
View System Information	System and Security

Add a Printer	Hardware and Sound
Change Date/Time	Clock, Language, and Region
Create User Account	User Accounts

9. Windows Accessories

9.1 What are Accessories?

Accessories are **built-in applications** included with Windows for performing basic tasks.

9.2 Common Accessories

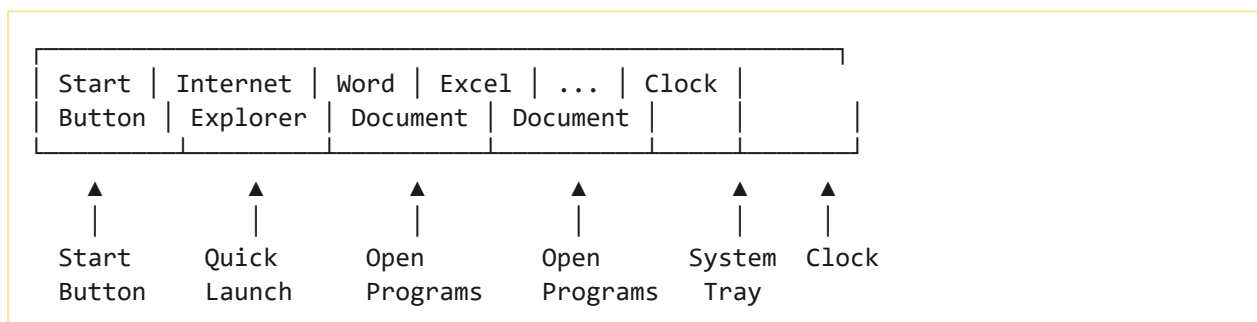
Accessory	Purpose	Launch Path
Notepad	Simple text editor	Start → All Programs → Accessories → Notepad
WordPad	Basic word processor	Start → All Programs → Accessories → WordPad
Paint	Graphics editor	Start → All Programs → Accessories → Paint
Calculator	Calculator application	Start → All Programs → Accessories → Calculator
Snipping Tool	Screen capture	Start → All Programs → Accessories → Snipping Tool
Sticky Notes	Digital sticky notes	Start → All Programs → Accessories → Sticky Notes
Sound Recorder	Audio recording	Start → All Programs → Accessories → Sound Recorder

10. Taskbar

10.1 What is Taskbar?

The Taskbar is a **horizontal bar** typically located at the bottom of the screen. It provides access to running programs and system controls.

10.2 Taskbar Components



10.3 Taskbar Features

Feature	Description
Start Button	Opens the Start Menu
Quick Launch	Shortcuts to frequently used programs
Program Buttons	Shows running applications
System Tray	Shows system icons and notifications
Clock	Displays current time and date
Notification Area	System notifications and alerts

10.4 Taskbar Functions

Function	Description
Switch Between Programs	Click program button
Launch Programs	Click Quick Launch icon
Show Desktop	Click Show Desktop button
Access Clock	Double-click clock for date/time
View Notifications	Click notification area

11. My Computer

11.1 What is My Computer?

My Computer (now called "This PC") is a **system folder** that provides access to all drives, files, and system resources.

11.2 My Computer Features

Feature	Description
Drives	Access to hard drives, CD/DVD drives, USB drives
System Information	Computer specifications
Manage	Disk management, device manager
Network Locations	Network drives and devices
Control Panel	Quick access to control panel

11.3 Accessing My Computer

Double-click My Computer on Desktop
or
Click Start → My Computer
or
Press Windows Key + E

12. Recycle Bin

12.1 What is Recycle Bin?

Recycle Bin is a **temporary storage location** for deleted files. Files can be restored if needed.

12.2 Recycle Bin Features

Feature	Description
Temporary Storage	Holds deleted files
Restore	Retrieve deleted files
Empty	Permanently delete files
Properties	Set storage limit

12.3 Recycle Bin Operations

Operation	How To
Restore File	Right-click file → Restore
Restore All	Recycle Bin → Restore All Items
Empty Recycle Bin	Right-click Recycle Bin → Empty
Delete Permanently	Shift + Delete (bypasses Recycle Bin)

13. DOS Command Summary

Command	Type	Purpose
DIR	Internal	List files
CD	Internal	Change directory
MD	Internal	Create directory
RD	Internal	Remove directory
COPY	Internal	Copy files
DEL	Internal	Delete files
REN	Internal	Rename files
TYPE	Internal	Display file
CLS	Internal	Clear screen
DATE	Internal	Set date
TIME	Internal	Set time
VER	Internal	Display version
VOL	Internal	Display volume
PROMPT	Internal	Change prompt
PATH	Internal	Set search path
FORMAT	External	Format disk
CHKDSK	External	Check disk
DISKCOPY	External	Copy floppy
XCOPY	External	Extended copy

TREE	External	Show directory tree
DELTREE	External	Delete directory tree
DOSKEY	External	Command history
ATTRIB	External	Change attributes
FIND	External	Find text
SORT	External	Sort data
FDISK	External	Partition disk

Wildcard Summary

Wildcard	Meaning	Example
*	Any number of characters	*.TXT
?	Exactly one character	FILE?.TXT

Redirection Operators

Operator	Purpose
>	Output to file (overwrite)
>>	Output to file (append)
<	Input from file
	Pipe output to next command

14. Questions

14.1 Short Answer Questions (2-5 Marks)

1. What is the difference between internal and external DOS commands?
2. What is the purpose of the DIR command? List its options.
3. Explain the wildcard characters used in DOS.
4. What are redirection operators? Give examples.
5. What is the purpose of the FORMAT command?
6. Explain the difference between CD and RD commands.
7. What is the function of the TREE command?
8. Explain the Recycle Bin in Windows.
9. What is the purpose of the Control Panel?
10. What are the benefits of Windows 2007?

14.2 Long Answer Questions (10 Marks)

1. Explain internal DOS commands with examples.
2. Explain external DOS commands with examples.
3. Explain the concept of files and directories with examples.
4. Explain wildcard characters and redirection operators in DOS.
5. Explain Windows 2007 features and benefits.
6. Explain the Control Panel and its functions.
7. Explain Accessories and their uses in Windows.
8. Explain the Taskbar and its components.

14.3 Objective Questions (1 Mark)

1. Which command shows file listings page by page?
(a) DIR (b) DIR /P (c) DIR /W (d) DIR /S

Answer: (b)

2. Which command creates a directory?
(a) RD (b) MD (c) CD (d) DEL

Answer: (b)

3. Wildcard for any number of characters:

- (a) ? (b) * (c) % (d) &

Answer: (b)

4. Which command clears the screen?

- (a) CLEAR (b) CLS (c) CRT (d) CLR

Answer: (b)

5. Recycle Bin stores:

- (a) Copied files (b) Deleted files (c) System files (d) Programs

Answer: (b)

6. Which command copies directories including subdirectories?

- (a) COPY (b) XCOPY (c) DISKCOPY (d) MCOPY

Answer: (b)

7. ATTRIB command is:

- (a) Internal (b) External (c) Both (d) None

Answer: (b)

8. The command to check disk errors is:

- (a) CHECK (b) CHKDSK (c) SCAN (d) DISK

Answer: (b)

End of Unit-I Notes